

A beginner's guide to Internet of Things



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The adoption of the internet of things (IoT) in Asia-Pacific has accelerated over the past few years, spurred by growing digitalisation across the region and the desire to improve business processes. According to a study by Omdia, 38.9 billion IoT devices are forecast to be in circulation in Asia-Pacific by 2030, with cellular IoT modules revenues expected to grow by 22% from 2021 to 2026. These devices are being used in a swathe of IoT applications, from tracking delivery trolleys in New Zealand to automating agriculture in Singapore. In this e-guide, learn more about IoT developments across the region and how to leverage the technology to drive business outcomes.

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Top 8 IoT applications and examples in business

Mary Pratt, Contributor

Enterprise spending on IoT projects is expected to ramp up this year.

Research firm IDC, in its "Worldwide Internet of Things Spending Guide," predicted a double-digit growth rate in 2021, with a compound annual growth rate of 11.3% from 2020 to 2024. IDC cited healthcare, insurance and education as leaders in IoT spending.

Researchers at Statista [predicted similarly high growth](#), saying the global IoT market for end-user offerings was expected to grow from \$212 billion in 2019 to \$1.6 trillion by 2025.

Statista also predicted there will be 75 billion IoT [devices in use](#) by 2023, generating [79.4 zettabytes of data](#).

Those IoT connections are spanning across the globe and throughout industries, as well as permeating individual homes, offices and vehicles, with the most prominent applications of IoT technologies falling into the following categories.

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1. Connected vehicles

Autonomous vehicles are one of the most notable examples of IoT in action. Self-driving cars and trucks use a slew of connected devices to safely navigate roadways in all sorts of traffic and weather conditions. The technologies in use include artificial intelligence (AI)-enabled cameras, motion sensors and onboard computers.

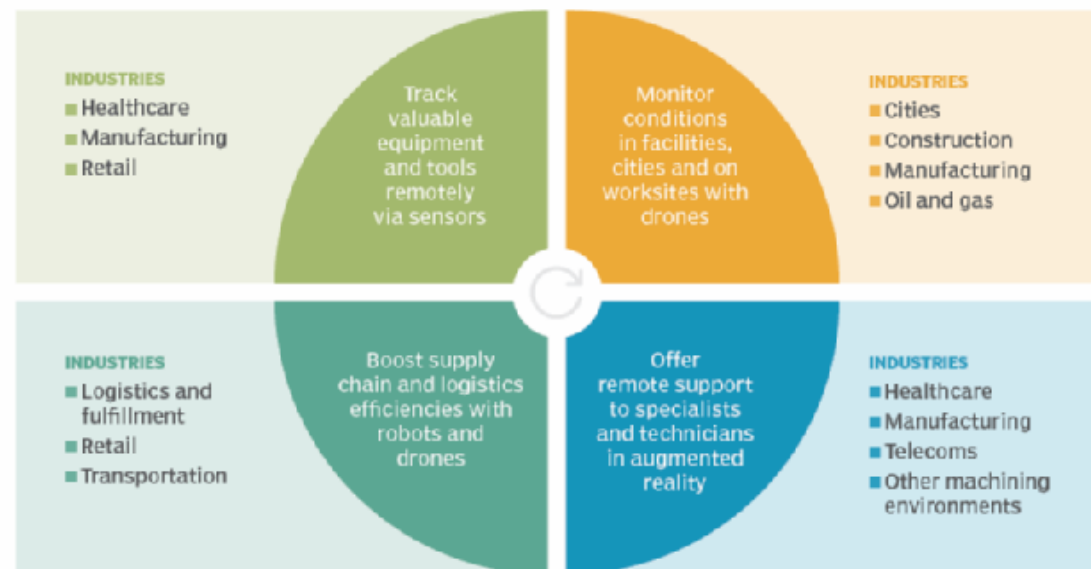
IoT connections also exist on conventional vehicles, with manufacturers installing connected devices to monitor performance and manage computerized systems.

Commercial fleets such as municipal buses and [corporate delivery trucks](#) are often fitted with additional IoT technologies, such as connected systems to monitor for safety issues. Personal cars and trucks can be fitted with similar technology, which frequently comes from insurance companies, that collects and transmits telemetry data to verify good driving habits.

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Industry IoT use cases for workforce distribution and automation



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2. Traffic management

Roadway infrastructure has become more connected in the past decade as well, with cameras, sensors, traffic light controls, parking meters and even smartphone traffic apps transmitting data that's then used to help avert traffic jams, prevent accidents and ensure smooth travel.

For example, [cameras detect and transmit data](#) about traffic volume to central management groups that can then analyze the information to determine whether, what and when mitigation steps must be taken.

Sensors on traffic signals can detect varying levels of light in the sky and adjust the brightness of the signals, helping ensure they're always visible to drivers.

Connected devices can be used to detect open parking spaces and transmit that information to kiosks or apps to alert drivers.

Monitors on bridges collect and transmit data for analysis about their structural health, alerting authorities to maintenance needs before there's any sort of failure or issue.

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3. Smart grids

[Utilities are also using IoT](#) to bring efficiency and resiliency to their energy grids.

Historically, energy flowed one way along the grid: from the generation site to the customer. However, connected devices now enable two-way communication along the entire energy supply chain: from generation through distribution to use, thereby improving the utilities' ability to move and manage it.

Utilities can take and analyze real-time data transmitted by connected devices to detect blackouts and redirect distribution, as well as respond to changes in energy demand and load.

Meanwhile, smart meters installed at individual homes and businesses provide information about both real-time use and historical usage patterns that customers and the utilities can analyze to identify ways to improve efficiency.

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4. Environmental monitoring

Connected devices can collect IoT data that indicates the health and quality of air, water and soil, as well as fisheries, forests and other natural habitats. They can also [collect weather and other environmental data](#).

As such, IoT delivers the ability to not only access significantly more real-time data about the environment at any given time and place, but it also enables a range of organizations in various industries to use that data to glean actionable insights.

Such information can help government agencies better monitor and even predict natural disasters, such as tornados, as well as better manage and protect land and wildlife populations. Companies can use this data to better limit their carbon footprint, more effectively document compliance with environmental regulations and/or more efficiently plan around weather conditions that affect their business.

5. Smart buildings and smart homes

Property owners are using the power of IoT to make all sorts of buildings smarter, meaning they're more energy-efficient, comfortable and convenient, as well as healthier and possibly safer, too.

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An [IoT ecosystem in a commercial building](#) could include monitoring of the HVAC infrastructure that uses real-time data and automation technologies to constantly measure and adjust the temperature for optimum energy efficiency and comfort. Meanwhile, cameras using AI could aid in crowd management to ensure public safety at events such as sold-out concerts.

On the homefront, consumers can install smart technologies, such as door locks, appliances, thermostats and smoke detectors, that help them with their everyday needs by, for example, coordinating temperature controls to the owners' schedules.

6. Smart cities

Smart cities are consolidating IoT deployments across many facets to give them a holistic view of what's happening in their jurisdictions.

As such, smart cities generally incorporate connected traffic management systems and their own smart buildings. They might incorporate private smart buildings, too. Smart cities might also tie into smart grids and use environmental monitoring to create an even larger IoT ecosystem that provides real-time views of the various elements that affect life in their municipalities.

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Similar to smaller, more confined IoT deployments, the objective with smart cities is to collect real-time data for analysis that provides insights that municipal officials can then use for better decision-making and/or automated controls to [yield more efficient, effective, resilient and safer communities](#). Case in point: Copenhagen, the capital of Denmark, is using IoT technologies to reach its goal of being a carbon-neutral city by 2025.

7. Supply chain management

Supply chain management has been undergoing a modernization, thanks to low-power sensors, GPS and other tracking technologies that pinpoint assets as they move along a supply chain. Such information lets managers both more effectively plan and more confidently reassure stakeholders about the location of items shipped or received.

That visibility is beneficial, but it's only the start of the [value proposition that IoT brings to this discipline](#). IoT technologies can also monitor and manage delivery requirements, for example, measuring and maintaining a specified temperature throughout transport to ensure quality and safety controls. Additionally, back-end analytics capabilities can use IoT-generated data to determine supply chain improvements, such as more efficient routes or shipping times.

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8. Industrial, agricultural and commercial management

IoT has numerous applications in industrial and commercial settings, enabling everything from predictive maintenance to improved security at facilities to smart agriculture. These wide-ranging use cases employ an equally expansive list of IoT technologies.

A manufacturer might use machine-to-machine connected devices as part of an [industrial IoT](#) deployment to more accurately map workloads. A factory could track wear and tear on equipment to schedule preventive maintenance at an optimal time. Companies can use employee badges or wearable devices embedded with RFID chips to manage and control physical access to their facilities. And farmers can opt for locational technologies integrated with environmental monitors and their field equipment to both automate and maximize their seed allocations.

Enterprise benefits

Although there are some industry-specific IoT use cases, many of the most common deployments involving connected technologies transcend any particular vertical and can be found across a wide range of organizations.

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The [benefits that come with those deployments](#) similarly cross industries. Typical ROI includes the following:

- lower operational costs;
- enhanced productivity;
- optimized output;
- increased security;
- improved safety;
- better quality control; and
- heightened understanding and visibility into real-world activities.

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Top advantages and disadvantages of IoT in business

Julia Borgini, Contributor

The global IoT market is on pace to surpass [\\$2.4 trillion annually by 2027](#), and more than 41 billion devices will be connected to the internet by 2027. IoT offers many benefits to businesses, including decreased operating costs, new consumer insights and opportunities to optimize business operations. It's not always a positive situation, though. Some businesses might only see the disadvantages of IoT devices and be hesitant to use them.

What are the advantages of IoT in business?

1. Lower operating costs

A fleet of [IoT devices](#) can be cost-effective because it helps businesses optimize their workflows and lower operating costs by providing real-time information. Devices can proactively advise of their status so staff can schedule the maintenance before it can affect production. They can be

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integrated into larger systems to optimize operational efficiency and help reduce costs. For example, smart building systems can track, monitor and control HVAC systems to monitor building usage and adjust them to take advantage of lower time-of-use costs, which saves money.

2. Increased productivity and workplace safety

IoT devices can manage, monitor and alert staff of changes in processes or productivity, helping them make smarter decisions about work. Automaker Ford is using [specialized IoT technology and body tracking sensor technology](#) to protect workers from excessive physical stress and to optimize their work. Engineers and ergonomists use the data to optimize each workstation to enable more efficient movement and help workers avoid injuries. Through this novel implementation of IoT in Ford's employee safety program, it's been able to [reduce assembly line injury rates by 70%](#).

3. Better customer experiences

IoT devices help businesses track, monitor, uncover and analyze customer data faster than before. Businesses can predict shifts or trends in customer behavior before they happen. Advanced IoT technology can enhance the

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[customer experience](#) by personalizing it based on past experiences. Think of the location trackers on shipping vehicles or personalized coupons offered through a mobile app on customers' smart devices when they enter a store or business. IoT devices can help businesses gather, transmit and analyze the personal data they have on customers, helping create a superior customer experience that engages them at a deeper level and increases customer loyalty.

4. More business insights

IoT devices help organizations gather data to identify insights about their business, both internally and externally. Retail stores use beacon technology and other IoT devices to redesign their stores based on real-time traffic patterns. Logistics firms can use internet-connected IoT devices to align delivery locations and schedules that make the most efficient use of vehicles and employees.

Businesses that use IoT to drive modernization throughout their organization reduce their time to market for new products or services and amplify their ROI. They'll add value to the business faster and more efficiently because more actionable data from the devices is easily accessible.

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What are the disadvantages of IoT in business?

1. Security and privacy

Keeping the data gathered and transmitted by IoT devices safe is challenging, as they evolve and expand in use. Although cybersecurity is a high priority, IoT devices aren't always included in the strategy. [Devices must be protected](#) from physical tampering, internet-based software attacks, network-based attacks and hardware-based attacks.

Data privacy is another concern, especially because IoT devices are being used in more sensitive industries, such as healthcare and finance. Information privacy laws are coming into effect globally, too, meaning that not only does it make good business sense to protect data, but businesses are legally required to do so.

[Integrating encryption](#) and security protocols with IoT devices can be difficult with a large fleet of devices. The cost in time, effort and money to do it on all devices might be prohibitive, so some businesses might use inadequate platforms because they're cheap or forego it altogether. All it takes is one breach for a business to learn a tough lesson.

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2. Technical complexity

Though it might seem like IoT devices are performing simple tasks, such as counting entry swipes at a secure door, there's a lot of complex technology involved in creating them. Plus, if they're providing essential data to another workflow or system, they could negatively affect everything connected to it. Miscounting the number of swipes at the door isn't a big deal, but if another device confuses temperature data with entry swipe data, it can be catastrophic. And the error isn't always easy to fix.

There can be a big learning curve in deploying IoT devices. It makes sense to develop a strategy on how and why to deploy them before purchasing them. That way, you can be assured they're working as intended, and you can support them.

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IoT advantages and disadvantages



Advantages

Lower operating costs

Higher employee productivity

Better customer experiences

New consumer insights



Disadvantages

Security and privacy issues

Lack of technical knowledge

Internet and power connectivity dependence

Time-consuming and expensive to implement

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3. Connectivity and power dependence

Many devices have a dependency on the internet and continuous power to function properly. When either goes down, so does the device and anything connected to it. Given how intertwined IoT devices are with today's businesses, everything can grind to a halt when they're down.

Businesses must understand how outages will affect their devices to plan proactively for outages -- because an [outage will happen at some point](#). Troubleshooting and incident management processes can help alleviate that, as can ensuring employees know what to do when devices are down.

4. Integration

There's currently no consensus regarding IoT protocols and standards, so devices produced by different manufacturers might not work with existing technology. Each one might require different configurations and hardware connections, making it hard to deploy efficiently.

The business must understand network needs to prepare for any required customization. That also means planning for extra time with device deployments to handle any troubleshooting or related tasks that might arise.

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5. Higher costs (time and money)

Deploying IoT devices often comes with high time and money investment requirements. There's the number of devices to be purchased and configured, staff to install them, others to integrate them into the network and support calls to the manufacturer for help. If they're all going into a single location, businesses can make up that investment quickly. If the business is spreading them out, expect the cost to rise exponentially.

By planning the [deployment budget and strategy](#) before buying, organizations can eliminate many potential obstacles they would otherwise encounter.

Summary of advantages and disadvantages of IoT

IoT devices can be useful to businesses today but only if they know what they're getting into. And these are just some of the benefits and disadvantages. With a little planning, businesses will have a better idea of what they need and how IoT devices can help.

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How APAC organisations can harness the power of IoT

Stephen Withers, Journalist

The adoption of the internet of things (IoT) in Asia-Pacific has accelerated over the past few years, spurred by growing digitisation across the region and the desire to improve business processes.

According to a study by Omdia, 38.9 billion IoT devices are expected to be in circulation in Asia-Pacific by 2030, with cellular IoT modules revenues expected to grow by 22% from 2021 to 2026. These devices are being used in a swathe of IoT applications, from tracking delivery trolleys in New Zealand to automating agriculture in Singapore.

In partnership with General Assembly, Computer Weekly's editor in chief for Asia-Pacific, Aaron Tan, moderated a panel discussion on the opportunities and challenges with IoT and how the technology is being used in different industries.

The panellists were Emily Bobis, founder of road intelligence company Compass IoT; Ram Kuppusamy, CEO and founder of digital aerospace company Space Angel; and Kien Hoong Fong, senior manager of data science at chipmaker Micron.

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Use cases and considerations

IoT is “a lot broader than most people think, and a lot more accessible,” observed Bobis.

Examples include connected cars, fridges, utility meters and even smartphones. The data collected from such devices is being used in various ways, such as providing better routing of trucks in the face of road closures following major floods, the delivery of real-time updates, and warnings of possible water leaks.

And in Australia, toll road operator Transurban has been able to use data from vehicles about swerving and braking to halve the number of crashes on one of Sydney’s major roads, she said.

Perhaps more prosaically, Micron uses IoT technologies to monitor the operation of equipment within its factories. Fong said vibration sensors can help determine whether fans are operating correctly or if they will soon need to be replaced, and cameras monitor robotic equipment to ensure that parts are being placed correctly and the tools are in good order.

In considering use cases for IoT, Kuppusamy pointed out the importance of understanding the business value – or indeed any other type of value – of a proposed IoT project. “It’s not about monitoring something or sensing something, it’s contributing to a tangible outcome” such as bettering a

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community, human lives or the environment, or helping achieve a business goal,” he said.

Organisations considering IoT implementations should also be aware of two particular barriers to adoption that are human rather than technological, Bobis noted.

First, there is still a need for more education about IoT, as “it can be a bit nebulous and a bit pretentious”, and that can put people off finding out more about IoT in general or specific IoT solutions.

Second, “innovation does not equal acceptance”. If something has been done in a particular way for decades, “you can have this amazing solution which does provide value”, but it won’t necessarily be accepted, she said.

“My favourite example of this is seat belts. When they first came out, people were so against the idea,” she said, adding that this was to the extent of arguing that you were more likely to die in a crash if you were wearing a seat belt – it took 25-30 years before they became mandatory.

So, organisations will need to adopt an innovation culture that allows risk taking, as people need to be confident that they won’t lose their job if they try something and it fails, Bobis said. Kuppusamy agreed but pointed out the importance of managing failures when they do occur.

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Challenges with IoT projects

Given that IoT is still a relatively new field, Kuppusamy's warning that "there will be some challenges" shouldn't come as a surprise. Those challenges can be encountered in various aspects of a project.

An obvious challenge is data volume. A sensor may be able to take measurements every second, but the communication and storage costs may outweigh the value of such finely grained data. And so, it might be more sensible to aggregate the data in some way, suggested Kuppusamy, such as averaging it over longer periods, or to simply take readings less frequently. However, it is sometimes important to capture data in real-time to detect crucial events when they occur.

There are also challenges associated with the environment in which sensors are deployed. For example, Kuppusamy said soil types could be an issue for a project aimed at collecting rainfall and river flow data at mining sites. This is because the ground is usually soft in salt mining areas, but very hard at iron ore sites, so securing the required sensors is non-trivial.

In remote areas, IoT implementations typically call for satellite communications as well as hardened devices that can function reliably over long periods without human attention. That reliability may also be required in urban areas for economic reasons, such as the cost of replacing batteries.

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As such, Kuppusamy suggested taking a value perspective. A \$10,000 device is unlikely to be justifiable for most projects, no matter how rugged or durable it is. On the other hand, having to manage devices, whether it is repairing them or changing batteries, is also expensive. In most cases, better results come from installing more sensors rather than better sensors.

Privacy and security are also important issues that do not always get the attention they deserve. Organisations are becoming more security conscious, said Tan, and it is becoming normal to assume that something may go wrong, so steps should be taken to prevent and mitigate security breaches.

“Talking about it is really important,” he observed, noting that part of the problem is that manufacturers may not be inclined to spend the money needed to secure cheap devices – although Singapore is incentivising IoT manufacturers to adopt a security-by-design approach.

With application programming interfaces (APIs) commonly used by IoT applications, the fewer APIs used and the simpler they are, the easier it is to secure them, suggested Kuppusamy. He recommended keeping things simple, and then taking steps to protect the simplest and most vulnerable APIs.

Another issue arises when actuators – rather than just sensors – are added to the mix. “If you go to a restaurant and there are robots serving food, the

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question is, if something goes wrong and the robot pours hot soup on somebody, who takes the liability?" asked Kuppusamy. And if your home security is based on IoT, "the last thing you want is to be locked out of your own home".

More generally, there is the question of whether an IoT system should be able to respond autonomously, or if there should be a human in the loop. "If your confidence levels are 90% and above, the capability of your solution is pretty good. But if your confidence level is at 50%, then obviously the human aspects to provide the contextual information is quite key," he said.

Industry landscape

Tan noted that the IoT landscape is highly fragmented, with suppliers playing in different parts of the IoT technology stack.

Kuppusamy pointed out that IoT suppliers may need to work on projects alongside their competitors if they do not have the necessary expertise in all aspects of the technology including devices, networks and applications. But as Tan pointed out, there is a tendency for IoT suppliers to expand their technology stacks over time.

IoT should also be regarded as an ecosystem, according to Fong, one that requires expertise in hardware, software, cloud systems, analytics and mathematical models and security. "It's a team effort," agreed Tan.

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That said, Kuppusamy stressed the importance for IoT technology providers to specialise in an area or industry: “There is a temptation to spread your net widely as the same sensor can be used for different applications in different industries, but you need to be able to service your customers effectively. Each project needs to help move the company in the direction it has set.”

Future developments

Not surprisingly given the companies they work for, the panellists see different opportunities in the IoT market in the next five to 10 years.

“I think one of the coolest things that’s going to happen in IoT is for transport planning applications around decarbonisation,” said Bobis, helping to answer questions including: ‘How can we get people to get an electric vehicle instead?’, ‘How can we put in enough infrastructure so that people who buy these vehicles can charge them on the street?’, ‘If everyone’s trying to charge their car at the same time, what is that going to do to the electricity grid?’ and ‘What about the bidirectional transfer of energy between the car and the grid?’.

Kuppusamy, too, sees opportunities in the utilities sector, and Space Angel is working with both the electricity and water industries.

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Fong's team is doing a lot of work in the sustainability area, collecting data about the use of water and other consumables, and then developing models to reduce that consumption in manufacturing plants. Among the questions being addressed are why different tools carrying out the same processes are consuming different amounts of water, he said.

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How Bosch is driving Industry 4.0 in India

Pratima Harigunani, Journalist

Bosch is one of India's earliest industrial pioneers, powering industries such as automotive and energy since it established its presence in the country a century ago.

In 1951, it set up its Indian manufacturing operations, which have been strengthened over the years to include 18 manufacturing sites and seven development and application centres.

More recently, the German conglomerate has localised advanced automotive technologies to support [India's desire to be more self-reliant and resilient](#) and is supporting the development of [Industry 4.0](#) in India by creating agile, future-ready plants and integrating local suppliers into its ecosystem.

Take its Bidadi plant, for example. In the past two years, the plant has not only become a low-cost manufacturing facility, but through a flexible labour model, it has been able to secure more temporary manpower while implementing advanced analytics and [artificial intelligence](#) (AI) to detect manufacturing defects.

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Its Nashik plant, on the other hand, has focused on behaviour-based safety to reduce first-aid cases and near-miss accidents, while the Naganathapura plant is implementing low-cost automation to improve operational efficiency.

Underpinning Bosch India's Industry 4.0 efforts are its software capabilities spanning [industrial IoT \(internet of things\)](#), [quantum computing](#) and AI.

Initially, the company teamed up with Indian companies with expertise in software development, but it later formed Bosch Global Software Technologies (BGSW) to tap India's vast technology talent.

Vadiraj Krishnamurthy, chief technology officer and vice-president of BGSW, said the company is now looking to strengthen its research and development (R&D) in areas such as [digital twins](#), connected mobility and supply chain automation. It has started a hackers' garage and an innovation fund to nurture and incubate promising ideas.

On the cutting edge of things, Bosch India has concrete plans to tap quantum computing for materials simulations.

Krishnamurthy said the company has had years of experience in materials simulations using conventional computers, along with the ability to mass-produce high-end technology products. This includes electric vehicles, which could be built with new materials designed by quantum computers

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that improve energy efficiency and reduce consumption of natural resources, he added.

Bosch India has also leveraged IoT and AI to build an [artificial intelligence of things \(AIoT\)](#) platform, which is being showcased at its campus headquarters in Bangalore, with people remaining at the centre of its AI efforts.

Krishnamurthy said: “The differentiating factor between AI and human intelligence is the human element of empathy. Our approach is to close this gap so that [humans and technology can work seamlessly](#).”

“We are running research projects on human compatible machines or ‘internet of behaviour’, which mainly focus on understanding human emotions along with sensing the environment through classic IoT to offer more relevant insights in our connected products.”

Meanwhile, like many other technology and industrial firms, Bosch India is driving efforts to [manage its carbon footprint](#). At the Bidadi site, various measures to reduce energy consumption have been implemented using a smart energy management system. This has enabled the plant to achieve 680MWh of energy savings in 2021.

On how the company is measuring the impact of its Industry 4.0 initiatives, Krishnamurthy cited two dimensions: how its employees are using digital

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technologies to work effectively and innovate, and how its customer and partners are experiencing its products and services.

“The way we look at measuring the impact is based on our digital [maturity model](#) which encompasses different elements like process optimisation, digital adaption and impact creation,” he said. “In the next two to three years, our aim is to jump by at least two to three steps ahead in our maturity model.”

Last month, Bosch announced that it will invest €10bn by 2025 in digitisation and connectivity. Two-thirds of that will go towards developing and expanding promising new technologies, with a focus on sustainability, mobility and Industry 4.0.

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Thinxtra teams up with Hong Kong's LBS on smart hygiene

Aaron Tan, Executive Editor, APAC

Australian [internet of things](#) (IoT) technology supplier Thinxtra has teamed up with Hong Kong's LBS Smart Technology to deliver smart hygiene applications to improve public health and safety for residential and commercial venues in the Chinese territory.

Under the partnership, LBS Smart Technology, a new subsidiary of LBS Group which provides hygiene management and purification technologies, will use Thinxtra's battery-powered IoT devices and [low-power wide area network](#) (LPWAN) in Hong Kong to monitor a range of hygiene and environmental indicators.

These include air quality, temperature, humidity and carbon dioxide (CO2) levels for a healthier environment, as well as rodent populations for more effective and sustainable management of rodent activities.

LBS Smart Technology will also use Thinxtra's devices to monitor water leaks and usage patterns of restrooms along with water quality to detect water temperature, pH levels and conductivity to ensure water quality is up to standard.

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Kong Chun Shak, CEO of LBS Smart Technology, said the company's partnership with Thinextra is an important step in making residential and public venues throughout Hong Kong even healthier and safer for all residents and visitors.

"We can take our services to the next level by measuring and analysing valuable data so that facility managers can monitor and track everything from indoor CO2 levels to reduce the risk of Covid-19 transmission, right through to the cleanliness of spaces to ensure better experiences for the community – from anywhere, at any time, and at scale," he added.

Thinextra CEO Nicholas Lambrou noted that the ability to monitor the condition of indoor environments remotely and around the clock is critical to maintain high-quality and risk-free facilities.

He added that the partnership will enable Hong Kong's offices, hospitals, universities and restaurants to be safer, more comfortable and convenient as property operators capitalise on the insights of real-time data, rather than relying on guesswork to keep venues in good shape.

"Working in partnership with LBS Smart Technology allows Thinextra to continue exploring the applications that will create value and drive efficiencies to businesses in Hong Kong," he said.

According to IDC, the IoT market in Asia-Pacific will continue to grow by 9.1% this year, accelerating from 6.9% in 2021.

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Although headwinds such as [semiconductor shortages](#), [supply chain disruptions](#) and rising inflation have dampened growth, the analyst firm noted that rising demand for remote operations, better network coverage, the deployment of commercial 5G and testbeds and government-led IoT initiatives are driving IoT adoption in the region.

China continues to account for the lion's share of the region's IoT market, followed by South Korea and India, while Singapore and Hong Kong have seen the fastest growth in IoT spending.

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NZ glassmaker taps IoT to track delivery trolleys

Aaron Tan, Executive Editor, APAC

New Zealand's Architectural Glass Products (AGP) is tapping [internet of things](#) (IoT) technology to track its products and delivery trolleys in real time and minimise asset losses.

The glass manufacturer has fitted tracking devices on more than 1,750 purpose-built delivery trolleys, which communicate over a Sigfox-based [low-power wide-area network](#) operated by Thinextra.

AGP spokesperson Gene Sanford said that the IoT deployment has proved central to the company's ability to deliver on its four-day order-to-delivery service.

"Rather than resorting to the time-consuming task of counting assets on a map with a basic GPS solution, we have a dashboard and data that inform us which customer has the trolleys and how long they have had them, enabling us to quickly take action to maintain our high service levels," Sanford said.

"Our customers' cash flow can be improved significantly knowing their supply will be delivered on time; getting our assets back quickly to fulfil the

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next set of orders helps mitigate any knock-on effect that might impact them,” he added.

The IoT devices are supplied by Pollin8, a local technology service provider. The devices are fixed to every AGP delivery trolley and record location data at all times, including at places where GPS signals are unavailable.

Pollin8 CEO Nick Pickering said that while GPS-based tracking solutions have been available for powered vehicles for decades, IoT technology makes it possible to track non-powered assets at scale, using devices with long battery life.

“That means we can provide AGP with not only around-the-clock tracking, but tailor the solution to its unique business requirements now and as its business expands in the future,” he added.

AGP claimed that the IoT deployment from Pollin8 and Thinextra was “extremely cost-effective”, making it suited to its plans for continued expansion. The IoT devices connect to Thinextra’s network easily without the need to build or maintain any communications infrastructure.

Thinextra CEO Nicholas Lambrou noted that AGP has built a “highly sophisticated operation underscored by incredible efficiency, which has catapulted the company to the pinnacle of glass supply in New Zealand”.

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“By bringing IoT into the mix from a very early stage, AGP gained granular visibility over the assets responsible for safely transporting its glass products to a fast-growing network of customers, avoiding unnecessary losses, and maintaining its stringent customer service level agreements.”

Besides AGP, which runs a four-hectare factory in Hautapu on Te Ika-a-Māui on North Island, New Zealand community housing provider CORT is also [harnessing IoT to improve the indoor air quality and ventilation of its premises](#).

Working with local technology company Tether, it is deploying sensors and analytics software, enabling it to monitor temperature, humidity, carbon dioxide levels, light and noise, among other environmental conditions. The sensors are also connected to Thinextra's Sigfox network.

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How IoT and machine learning are automating agriculture

Aaron Tan, Executive Editor, APAC

A new generation of farmers is turning to technologies such as the [internet of things](#) (IoT) and [machine learning](#) to automate agricultural production, alleviating the need to toil on the land while keeping a watchful eye on their crops.

Joining their ranks is Phoebe Xie, director and co-founder of Singapore-based agrotechnology startup AbyFarm. Teaming up with technology service provider SPTel, Xie is building a smart hydroponics farm in a greenhouse that uses a plethora of IoT sensors, including video cameras, to keep the farm humming around the clock.

“To run a self-regulating farm at optimal temperature with optimal water and nutrient supply, and to control the quality and taste of vegetables and fruits, we need IoT sensors,” she said, adding that the farm and its sensors collect and monitor thousands of data points, including humidity and temperature.

With the data and in certain environmental conditions, processes and actions are automatically triggered to protect crops from the elements. For instance, if the temperature or humidity gets too high, fans, water curtains and roof shades are activated in the greenhouse.

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Xie said the smart farm, located at a rooftop carpark in Singapore, is also equipped with sensors that monitor the pH and electrical conductivity levels of water. Among the sensors, which are connected to a [Lora low-power wide area network](#), is a dozer that automatically releases acidic or alkaline nutrients to maintain optimal pH levels.

Whereas the farmers of old had to physically inspect their crops, AbyFarm's agronomists and farmers can do so remotely using video cameras and [image recognition](#) technology to identify crops that might be succumbing to disease.

"If the image recognition finds that a plant is likely to be sick, the farmer will be alerted to go onsite to check how the plant is right now," said Xie, adding that agronomists will also advise farmers on taking corrective actions to prevent disease or restore the health of their crops.

Like any machine learning system, AbyFarm's algorithms get smarter over time. Each time a crop is affected by disease, data about the occurrence and corrective action is fed into the system, enabling its algorithms to formulate solutions for other farmers with similar crop issues – without consulting an agronomist.

With that, said Xie, even those who are new to the field can use the system to guide them on treating diseased crops, which can be isolated from other crops to prevent cross-infections.

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AbyFarm's automated farming system, which is hosted on SPTel's private cloud, can also advise farmers on the best time to transplant their crops after germination and harvest them later for sale, said Xie.

Heng Kwee Tong, vice-president for engineering and customer solutions at SPTel, said the company has built up a [software-defined platform](#) to help companies like AbyFarm overcome the challenges of deploying applications such as IoT.

"The common struggle that companies like AbyFarm face with building an IoT solution is that it is quite intensive in terms of ICT infrastructure," said Heng. "You've got to find the radio connectivity like Lora, subscribe to a service provider and connect your sensors to a sensor collector."

"We make it easy by investing in a platform and because we are a service provider with edge hub assets for deploying radio connectivity, all AbyFarm needs to do is to bring their sensors onboard."

Going forward, Heng said SPTel is looking to support next-generation [video analytics](#) capabilities through [edge cloud services](#) that can be used to crunch workloads closer to where they reside, reducing latency.

AbyFarm was one of six companies to win contracts from the Singapore Food Agency last year to build urban farms at nine rooftop carparks in public housing estates across Singapore.

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The goal is to collectively produce about 1,600 tonnes of vegetables a year in a bid by the city-state – where less than 1% of land is used for agriculture – to become more self-sufficient in meeting the nutritional needs of its people.

That could well be achievable with technology that makes farming [more efficient, scalable and less labour-intensive](#) than before. “We are willing to train and educate the next generation in agrotechnology, because now you don’t need 10 farmers to take care of one farm,” said Xie. “It’s just looking at the dashboard and, with the alerts, doing what is required.”

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